



EXPLORING AUGMENTED REALITY (AR) AND VIRTUAL REALITY (VR): CONCEPTS, ADVANCES AND APPLICATIONS



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Abstract: This report describes an in-depth analysis of Augmented Reality (AR) and Virtual Reality (VR), delving into their historical developments, fundamental principles, technological progress, and widespread applications. It emphasizes the predominant trends influencing the evolution of AR and VR, particularly developments in hardware, software, and methods of user engagement. Additionally, the paper investigates the incorporation of these technologies across multiple sectors, including healthcare, education, entertainment, and industry, demonstrating their transformative effects and future possibilities.

Keywords: Augmented reality, Virtual reality, Evolution, Hardware, Software, Application.

Introduction

Augmented Reality (AR) is a technological innovation that integrates digital components with the physical environment, resulting in an engaging and immersive experience. This technology combines real-time digital visuals, sounds, and videos with the actual world to enhance user experiences. The spectrum of AR applications ranges from basic images and text to intricate animations and interactive simulations, allowing users to engage with both the physical and digital realms simultaneously (Nakul *et al.*, 2024). AR presents opportunities for user interaction and meaningful learning regarding the operations of 3D-registered elements within AR systems. However, despite these advancements, maintaining the quality of AR experiences continues to be a complex and evolving challenge. The quality of augmented reality (AR) experiences is a nuanced concept that involves critical aspects such as semantic coherence, precision in spatial alignment, interaction responsiveness, and visual fidelity. Realizing the ideal quality of AR requires a deep understanding of how these factors interact, as well as the broader elements that shape user perception and satisfaction. To tackle this issue, contemporary literature has increasingly embraced conceptual models that provide organized frameworks for evaluating and enhancing AR quality (Mohammad *et al.*, 2024). With ongoing technological advancements, AR is projected to become more accessible, affordable, and powerful. The foundation of AR can be linked to earlier computing innovations, such as smartphones and wearable technology.

Virtual Reality (VR) refers to a technology that can elicit specific behaviours in humans or organisms by employing artificial sensory stimulation, often without their conscious recognition of the interference. It creates a computer-generated simulation that immerses users in a three-dimensional (3D) interactive space (Zhang, 2022). Merriam (2023) characterizes VR as an artificial environment that is perceived through sensory inputs provided by a computer. Users interact with this environment using specialized electronic devices, such as headsets with screens or gloves equipped with sensors. The application of this technology has become increasingly popular in recent years (Kersten *et al.*, 2024). Objects that have ceased to exist or are only represented

in designs and blueprints can be visualized, providing a diverse array of representations that incorporate user presence within the artificial environment (Knoll and Stieglitz, 2022). Virtual reality (VR) facilitates the transfer of knowledge through interactive experiences within the VR framework (Kersten *et al.*, 2020). As technological advancements continue, the future of VR appears bright, with ongoing efforts to integrate artificial intelligence (AI), haptic feedback systems, and brain-computer interfaces (BCIs) that could significantly enhance the VR experience. It is anticipated that VR will transition from a specialized tool to a pervasive technology woven into everyday life.

AR and VR play a crucial role in shaping modern human society. These technologies have revolutionized our comprehension of the mind, body, and actions in relation to the environment, thereby impacting contemporary thought processes. The varied insights provided by AR and VR have allowed humanity to delve into a wider array of interests, promoting research and innovation. These technologies find applications in numerous fields, including industry, construction, architecture, medicine, and scientific research (Olajide *et al.*, 2017). Furthermore, the concepts of different realities are based on factors such as the level of immersion, degree of interaction, and the evolving nature of experiences (Palmarini *et al.*, 2018). As a result, this paper aims to conduct a systematic review of the literature to investigate the applications of AR and VR in human civilization. It will also evaluate their use cases across various sectors and examine the current advancements in the field.

History and Evolution of Augmented Reality

The history of Augmented Reality (AR) reflects a journey of technological evolution, beginning with foundational theories and experimental systems to its current status as a transformative technology across various industries (Rauschnabel and Ro, 2020). As AR continues to grow and evolve, it holds the potential to redefine how we interact with the world, providing new opportunities for education, entertainment, communication, and beyond (Kipper and Rampolla, 2019; Adeosun *et al.*, 2016). The ongoing advancements promise to make AR an integral part of our daily lives, blending the digital and physical spaces like never before. AR is an interactive experience that combines the real world and computer-generated 3D

contents (Dünser and Wiklund, 2020; Azuma, 2019). The overlaid sensory information can be constructive (i.e., additive to the natural environment), or destructive (i.e., masking of the natural environment). As such, it is one of the key technologies in the reality-virtuality continuum. Augmented Reality (AR) is a technology that overlays digital information onto the real world, enhancing the user's perception and interaction with their environment (Carmigniani and Furht, 2021). While the concept may seem relatively modern, the roots of AR can be traced back several decades. This paper explores the key milestones in the history and evolution of AR, from its foundational concepts to its current applications across diverse fields;

Early Foundations (1960s - 1990s)

The Sensorama (1960s): The origins of immersive experiences can be seen with Morton Heilig's development of the Sensorama in the 1960s. Although primarily a multi-sensory experience rather than interactive AR, the Sensorama combined visuals, sounds, vibrations, and scents to immerse users in a particular environment.

The First Head-Mounted Display (1968): Ivan Sutherland created the first head-mounted display (HMD), known as the "Sword of Damocles." This pioneering device projected simple wireframe graphics onto a user's field of vision. Although groundbreaking, it lacked practicality due to its weight and limited visual quality.

Virtual Fixtures and Early AR Systems (1980s): In the late 1980s, Louis Rosenberg developed the Virtual Fixtures system at the U.S. Air Force's Armstrong Labs. This system could enhance physical tasks by overlaying digital information, functioning as one of the first practical applications of AR.

Augmented Reality Architectural Interface (ARAI), (1992): Sutherland and Rosenberg's work laid the groundwork for real-world applications, with the (ARAI) enabling the visualization of virtual objects in real environments, especially in architectural design.

Development and Research (1990s - 2000s)

Academic and Industry Exploration (1990s): Interest in AR grew within academic and industrial research circles. Projects aimed at improving AR technologies addressed issues of tracking accuracy, user interface design, and real-time interaction, setting the stage for future developments.

Introduction of ARToolKit (1999): Hirokazu Kato developed ARToolKit, an open-source software library that became essential for creating AR applications. It enabled developers to create AR experiences by providing real-time tracking and rendering of virtual elements based on camera input.

The Rise of Mobile AR (2000s): As mobile technology advanced, researchers and developers began exploring AR on smartphones. Early frameworks allowed users to point their phones at real-world locations or objects to receive additional information, signifying the potential of AR in mobile environments.

Commercialization and Popularization (2000s - 2010s)

Introduction of AR Browsers (2009): The launch of AR browsers like Layar, Wikitude, and Junaio transformed mobile devices into AR platforms. These applications used GPS, compass, and camera features to provide location-based information, enhancing activities like tourism and navigation.

The Emergence of Social AR (2012): While applications in gaming and navigation flourished, social AR began to take shape, enabling users to share AR content and experiences on social media platforms, thus broadening the appeal of AR technologies.

Google Glass (2013): Google introduced Google Glass, a wearable device with an HMD that could display notifications, maps, and other information. Although the product faced criticism regarding privacy and functionality, it sparked significant interest in AR wearables.

The AR Boom (2016-Present)

Pokémon GO (2016): The release of "Pokémon GO" marked a seminal moment for AR, bringing the technology into mainstream culture. The game's success demonstrated AR's potential to engage users in the real world through a fun and interactive experience, blending digital gameplay with physical exploration.

ARKit and ARCore (2017): Recognizing the potential of AR, Apple launched ARKit in 2017, providing developers with tools to create AR applications for iOS devices. Google followed suit with ARCore for Android, both of which facilitate the integration of AR into existing mobile ecosystems.

Diverse Applications: Across healthcare, retail, education, and entertainment, AR technologies are being applied to improve experiences and outcomes. In healthcare, AR aids surgeries and medical training by overlaying critical patient information and anatomical visuals in the surgeon's field of view. In retail, AR allows customers to visualize products in their own spaces before purchase, thereby enhancing consumer confidence.

Future Trends in AR

The development of more sophisticated hardware, such as smart glasses and contact lenses, is anticipated to further enhance AR experiences. There is also ongoing research into improving display technology, making devices more lightweight, and increasing their functionality.

The convergence of AI and AR promises to create intelligent, context-aware applications capable of delivering personalized experiences based on user behaviour and preferences.

History and Evolution of Virtual reality (VR)

A rapidly developing area of technology is virtual reality (VR), which uses digital components such as sensors, screens to improve a 3D environment for users. Since virtual reality is completely immersive, that is, it presents a simulated environment that replaces the user's actual surroundings in enough detail for the user to engage with, using software that only exists virtually (Balasai *et al.*, 2022). VR was first mentioned back in the 60s, though its origin can be traced to the 19th Century when the first 360-degree art through panoramic murals first surfaced. After about one hundred years, precisely in 1062, the first form of VR emerged as a mechanical equipment, the Sensorama, created an immersive VR, using several senses. The system offered a multisensory practice of riding a motorcycle, including three-dimensional, full colour film together with sounds, smells, and the feeling of motion, as well as the sensation of wind on the viewer's face (Friena & Ott, 2023; Balasai *et al.*, 2022). Since then, virtual reality has changed and gotten more like the actual world in a number of ways. The growing computational power and the increasing completeness and adaptability of

human-computer interfaces have made ICT and VR synonymous.

Ivan Sutherland in Sutherland (1963) released Sketchpad, a computer application that utilised a tracked light pen and an x-y vector display to facilitate computer-aided drawing. Arguably, this was the first computer-connected interactive graphical user interface. The "ultimate display," according to Sutherland two years later, is "a room within which the computer can control the existence of matter" (Sutherland, 1968). "A chair that is displayed in such a room would be good enough to sit in." Displaying handcuffs in such a space would be restrictive, and displaying a bullet there would be lethal. Sutherland eventually developed the first Head Mounted Display system for interactive computer graphics with his pupil Bob Sproull. This technology produced binocular images that were suited to the moving head's orientation and position. The display was hung on a counterbalanced robotic arm, as seen in Figure 1, and ultrasonic transducers were employed to monitor the head's natural motion.

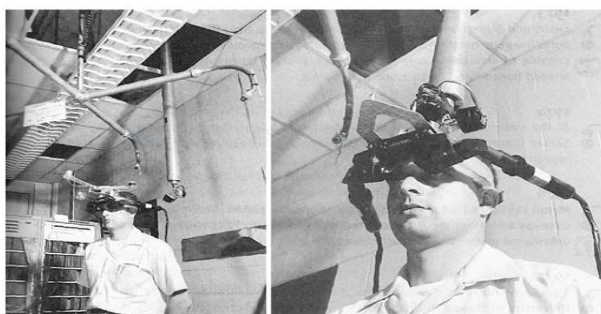


Figure 1: The First Head Mounted Display System (Basu *et al.*, 2012)

People were able to view a computer-generated virtual environment for the first time in computer graphics history. According to Sutherland, "make that (virtual) world in the window look real, sound real, feel real, and react realistically to the viewer's actions (Sutherland, 1968).

It is crucial to research the history of the technologies that have shaped the field of virtual reality in order to comprehend the present trends in this area. Examining the significant turning points that resulted in the development of VR technology makes it clear where many contemporary initiatives are coming from. The fundamental components of virtual reality have been there since 1980, but their implementation required robust image rendering. This pattern persisted until the advent of cellphones in the late 2000s. By 2011, there was a growing chance of experiencing fully untethered immersive virtual reality. This paved way for the various application areas of Virtual Reality being utilized presently and high-performance computers (Chen, 2022).

Structure of Augmented Reality Systems

The core elements of Augmented Reality systems include hardware, software, and applications that support the projection of digital information onto the real world, permitting real-time engagement with the virtual objects situated in the three-dimensional space of the augmented environment (Zhang, 2022; Knoll and Stieglitz, 2022).

AR System Hardware Components

The hardware components utilized in augmented reality (AR) systems include the following;

- (a) **Devices:** These encompass smartphones, tablets, smart glasses, and specialized headsets.
- (b) **Cameras:** Their function is to capture scenes from the real world.
- (c) **Sensors:** Components such as accelerometers, gyroscopes, and GPS are employed to monitor movement and determine location.
- (d) **Displays:** These are responsible for presenting augmented content, which can be superimposed onto the real environment.
- (e) **Network infrastructure:** This is essential for data access and transmission.
- (f) **Trackers and Sensors:** Instruments like GPS, compasses, and accelerometers are utilized for tracking and alignment purposes.
- (g) **Microprocessors:** These devices process data received from sensors and cameras, facilitating the generation of augmented content (Zhang, 2022).

AR System Software Components

The components of software employed in the development of augmented reality (AR) systems include:

- (a) **AR Applications:** These programs are designed to create and manage the augmented reality experience.
- (b) **Tracking and Rendering Engines:** These systems handle camera input, track user movements, and render virtual elements.
- (c) **3D Models and Textures:** These digital features are applied to the real-world environment.
- (d) **User Interface:** This interface facilitates user interaction with the AR application (Knoll and Stieglitz, 2022)

AR System Application Components

The elements that constitute AR systems are:

- (a) **Interactive Experience:** AR facilitates user interaction with their actual environment alongside virtual content.
- (b) **Real-time Integration:** AR integrates digital information with the physical world instantaneously.
- (c) **Enhanced Perception:** AR enhances a user's perception of reality by overlaying digital information (Knoll and Stieglitz, 2022).

Types of AR

AR systems finds applications in various field; however, they are categorised into three types of AR systems;

- (a) **location-based AR system:** is a location-aware system that superimposes virtual data onto real-world objects, leveraging a user's geographical position (such as GPS or Wi-Fi) to deliver contextually relevant information and experiences, independent of physical markers. The content of a location-based augmented reality (AR) system is customized to the user's specific location, enhancing its relevance and engagement. For instance, such a system could provide details about nearby landmarks when you direct your phone towards them or assist you in navigating a museum with digital displays and visuals. In essence, a location-based AR system enriches your environment by

making it more interactive and informative according to your location.

- (b) **Marker-based AR system:** This category of augmented reality (AR) systems employs predefined visual markers, such as QR codes or images, to serve as anchors for overlaying digital content onto the physical environment. These systems facilitate AR experiences by utilizing a camera to recognize these markers, subsequently displaying virtual elements in their appropriate locations. The operation of these AR systems involves detecting markers on real-world objects through camera input. Markers consist of unique patterns that can be easily recognized by the camera. Upon identifying a marker, digital information is superimposed onto the corresponding real-world object. The system then continues to monitor the position and orientation of the markers, ensuring the precise placement of the virtual content. For instance, when a user points their smartphone at a marker on a book cover, the application may present a 3D animation or additional information pertaining to that book. Marker-based AR systems are known for their accuracy, simplicity, and cost-effectiveness in implementation.

- (c) **Projection-based AR system:** This particular augmented reality (AR) system integrates projectors and vision sensors to present interactive graphics in a step-by-step manner on any work surface. The operation involves a projector that casts images, animations, or videos onto a physical surface, while sensors monitor the users' gestures and positions in relation to the projected virtual elements. Throughout the duration of use, the system dynamically adjusts the virtual content in response to any changes in the users' positions or gestures. For instance, a projector might display a map on a table, allowing users to interact with the map by touching the table at the locations where the digital content is shown. Although projector-based AR systems can create a powerful emotional impact through the lifelike representation of digital objects, they are constrained by the dimensions and shape of the surface, affected by ambient light, and necessitate intricate setup (Nakul *et al.*, 2024; Mohammad *et al.*, 2024).

Structure of VR Systems

Software and hardware are the two main subsystems that comprise a VR system. The hardware can be further separated into I/O devices and computer or VR engines, and the software can be further separated into databases and application software (Vu, 2022), as shown in Figure 2.

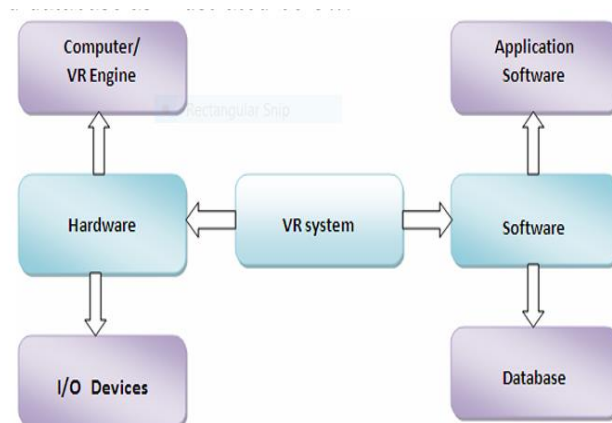


Figure2: Virtual Reality System (Bamodu and Ye, 2013)

System Hardware for Virtual Reality

Input devices, output devices, and the VR engine or computer system are the main parts of the hardware as depicted in Figure 3.



Figure 3: System Hardware for Virtual Reality (Bamodu and Ye, 2013)

Devices for input

The user interacts with the virtual world through the input devices. In order to offer the user with suitable responses through the output devices in real time, they transmit signals to the system regarding the user's action. They fall into four categories: voice devices, bio-controllers, tracking devices, and point input devices. Position sensors, often known as tracking devices, are used to track a user's position. These devices include data gloves, neural and bio or muscle controllers, electromagnetic, ultrasonic, optical, mechanical, and gyroscopic sensors (Dani & Rajit, 1998; Craig *et al.*, 2009).

VR Engine

The computer system or VR engine used in VR systems must be chosen based on the application's requirement. Data gloves, neural and bio or muscular controllers, and graphic display and image generation are some of the most crucial and time-consuming components of a virtual reality system (Vu, 2022). The selection of the VR engine is contingent upon the application domain, user, I/O devices, degree of immersion, and the necessary graphic output, as it is in charge of graphical model computation and generation, object rendering, lighting, mapping, texturing, simulation, and real-time display. In addition, the computer manages user interactions and acts as an interface for the I/O devices (Bamodu and Ye, 2013). When choosing a VR engine, one important consideration is the computer's processing capability, which is determined by how many senses (sound, haptic, graphical, etc.) can be produced in a specific length of time. The VR engine must generate real-time simulations at over 24 frames per second and recalculate the virtual environment around every 33ms (Burdea and Philippe, 2003).

Devices for output

In order to arouse the senses, the output devices receive feedback from the VR engine and transmit it to the users via the appropriate output devices. Based on the senses, output devices can be categorised as haptic (touch or force), visual (graphics), auditory (sound), olfactory (tasting). Smell and taste are still uncommon, however, the first three are commonly utilised in VR systems (Sherman and Craig, 2018). The HMD, which offers a greater degree of immersion, and the stereo display monitor are two potential popular choices for the graphics. The brain interprets the two separate perspectives that are generated by the HMD to construct a three-dimensional view of the virtual environment. Only the visual channel is more significant in virtual reality than audio or sound. To add realism to the VR application, 3D sound can be used to create various noises from various locations (Vu, 2022). The user can experience virtual items through the use of haptic. This can be accomplished by mechanical mechanisms or through electrical signals.

Tools and Software for Virtual Reality Systems

A suite of tools and software for creating, developing, and managing simulated environments as well as the database where the data is kept is known as virtual reality system software. The tools fall into two categories: development tools and modelling tools. Modelling tools for designing VR include Maya, Creator, and 3ds Max. Applications specific to engineering may make use of programs like CATIA, Pro/E, Solidworks, UG, and others. Tools for VR development have to be carefully chosen due to its complex and integrative technological requirements. Among the VR development tools used in the production of VR content are virtual world authoring tools, VR toolkits/software development kits (SDK), and application program interfaces (APIs). Nonetheless, it is common to find that a number of APIs, such the OpenGL optimiser and Java 3D API, are also utilized (Dani and Rajit, 1998).

VR System Operation

The basic operations of a VR system are:

Initialization: At this stage, the user holds the controls while wearing the HMD and tracking of the user's movements starts after the system initializes.

Tracking: The tracking system in the VR system sends data to the computer or console, which uses the information to update the virtual environment.

Rendering: The computer or console renders the virtual environment, taking into account the user's movements and interactions.

Display: To provide the user with a completely immersive experience, the HMD displays the produced virtual environment.

Feedback: The system receives input from the sensors, which enables it to modify the overall experience in real time (Steed and Slater 2008; Campos and Fernandes, 2015).

Applications of AR

Augmented reality can be useful in factually many aspects of life to enhance natural environments or situations and offers perceptually enriched experiences. With the help of advanced AR technologies, the information about the surrounding real world of the user becomes interactive and digitally manipulated (Dargan *et al.*, 2023; Wu *et al.*, 2013). Some selected areas of application of AR are explained as follows:

Training and Education

AR in education and training opens up new possibilities for how we learn and teach. Whether it's bringing textbooks to life, simulating real-world scenarios, or providing interactive, hands-on experiences, AR makes learning more immersive, engaging, and effective. As AR technology continues to evolve, its impact on education is likely to grow, offering increasingly powerful tools for teachers, students, and professionals alike. Dhaas, (2024). systematically examines AR applications in various academic fields, comparing them to traditional educational technologies and teaching methods, through a comprehensive review of research on the potential of AR in education. Their studies reported that AR exhibits significant potential to revolutionize education by transforming the learning process into a more active, effective, and meaningful experience. AR accomplishes this by providing advanced technology that allows users to interact with virtual and real-time applications, thereby immersing them in realistic educational scenarios. Across a spectrum of subjects, including Medicine, Chemistry, Mathematics, Physics, Geography, Biology, Astronomy, and History, AR offers advantages over conventional technology and traditional teaching methods. This research underscores the positive potential of AR in education and highlights its capacity to bridge the gap between technology and active, engaged learning. To fully harness these benefits, educators and researchers should address the identified limitations of AR and explore its further integration into educational practices.

Design and Architecture

AR is revolutionizing design and architecture by allowing professionals to visualize and interact with 3D models in real-world settings, improving collaboration, decision-making, and overall project efficiency. Whether for visualizing interiors, presenting design concepts to clients, or managing construction projects, AR tools are becoming invaluable in creating smarter, more effective design workflows. As AR technology continues to evolve, it will likely play an even bigger role in reshaping how architectural and design projects are planned, executed, and experienced. AR tools and software are transforming the workflow of designers and architects by enabling them to visualize their designs in real-world contexts before making physical changes. These tools are improving communication with clients, reducing errors, and enhancing creativity in the design process. AR is becoming a powerful part of the design toolkit and here are some examples IKEA Place, Hololens by Microsoft, Morpholio AR, SketchUp Viewer with AR, Augment, ARki, PlanGrid (by Autodesk), Vuforia, Fuzor, Magic Leap, etc. Alekhya, *et. al.*, (2023) conducted research that examined the transformative effect of augmented reality (AR) on the practice of architectural design, including the entire process from basic conceptualization through ultimate execution. The present investigation analyzes the utilization of augmented reality (AR) as an approach to enhance visualization, increase design accuracy, and ease client communication. The study takes into account advantages of augmented reality (AR) in facilitating the production of realistic and engaging design experiences addresses the challenges and obstacles linked to augmented reality (AR) technology, such equipment

standards and technical knowledge. The result provides a comprehensive examination of current uses of augmented reality (AR) in architecture, utilizing a combination of mathematical modeling and practical case studies. Also, it analyzes the potential of AR to reshape the next phase of architectural design.

Games

The real-time integration of game visual and audio elements with the user's environment is known as augmented reality gaming (AR gaming). Augmented reality gaming does not necessitate a separate room or confined area to create an immersive environment, unlike virtual reality gaming, which frequently necessitates a separate room or confined area. Mobile gaming systems, such as smartphones and tablets, are popular platforms for augmented reality (AR) games. Duan, (2018) systematically reviews the research status of vision-based augmented reality technology. First, it outlines the current research status and application of augmented reality technology; secondly, it elaborates the key technologies of augmented reality, including display technology, tracking registration technology, interaction technology, and calibration technology. Finally, according to the different working methods, All AR games are divided into two categories, basing on fixed patterns and basing on geographical location, giving a brief introduction to the AR games that stand out on the market.

Therapy and Rehabilitation

Naqvi *et al.*, (2024) work reflects the potential of the AR and VR in their capacity to create immersive, interactive environments that facilitate the recovery. It has also illustrated the ability to enhance the patient engagement and outcomes, especially in tackling the complex motor and cognitive rehabilitation needs. The combination of AR and VR with artificial intelligence (AI) technology has brought rehabilitation to the next level by enabling adaptive and responsive treatment programs provided through real-time feedback and predictive analytic methods. Having examines the many potential gains, drawbacks, and future directions from a different perspective which has improved cross-disciplinary operations. The result makes rehabilitation programs more engaging, providing realistic environments for better rehabilitation outcomes, allowing personalized therapy, and facilitating data collection for therapy optimization.

Applications of VR

VR has a wide range of current and emerging applications across different disciplines and sectors. these include training and education, design and architecture, therapy and rehabilitation, medicine, simulations, engineering, and entertainment (Hamad and Jia, 2022). Despite this growth, an in-depth knowledge of VR's applications and their roles in technological advancement, strengths and limitations across these diverse application domains is still developing (Hamad and Jia, 2022).

VR in Games and Entertainment

The gaming industry was among the first to embrace VR technology, and it remains a significant area of application. VR games offer immersive experiences, enhancing user engagement through interactive and spatial elements. Beyond gaming, VR is also transforming the broader entertainment industry, offering new ways to experience movies, concerts, and other forms of media. The technology's ability to create realistic and engaging simulations makes it ideal for entertainment purposes. Li,

(2024) explores the application of Virtual Reality (VR) games as tools for preserving and promoting cultural heritage, addressing the problem of limited accessibility and engagement with historical artifacts and traditions. Using case studies such as the "Rome Reborn" project and the British Museum's VR initiatives, the research highlights how immersive, interactive, and educational experiences improve user engagement and foster a deeper appreciation for cultural heritage. However, it identifies gaps in historical accuracy, cost-effective implementation, and specialized resources for VR game development, while suggesting future research focus on integrating VR with emerging technologies like artificial intelligence and big data for enhanced personalization and accessibility.

VR in Training and Education

Virtual Reality creates exclusive opportunities for training and education across various disciplines. It allows students and trainees to experience lifelike simulations and interact with systems in a motivating and engaging way. The technology is particularly valuable in fields where practical experience is needed, such as medicine, architecture and engineering. Long *et al.*, (2024) conducted a comparative analysis and in-depth investigation on the concept of the application and effect of virtual reality (VR) technology in vocational education practical training, addressing challenges like high costs, safety risks, and limitations of traditional methods. Through a three-month experimental study involving 200 students, using surveys, interviews, and skill evaluations, it shows VR significantly enhances theoretical knowledge, practical skills, and student satisfaction while reducing costs and risks. Despite its benefits, challenges such as equipment costs and insufficient specialized teaching resources highlight a gap in developing accessible and scalable VR-based teaching systems. The author suggested that future research could explore the integrated application of virtual reality technology with emerging fields such as artificial intelligence and big data, aiming to further enhance personalized learning experiences and teaching effectiveness in vocational education.

Chen *et al.*, (2020) compared the educational effectiveness of a virtual reality (VR) skull model with traditional cadaveric skulls and atlases in teaching anatomy to medical students. The study conducted a mixed-methods approach, dividing 73 students into three groups: VR, cadaver, and atlas, and assessed their learning outcomes through pre- and post-intervention tests and perception surveys. The results showed that all groups significantly improved their test scores post-intervention, with no statistically significant differences across the groups, but the VR and cadaver groups provided more positive feedback on their learning experience compared to the atlas group. The author suggested that future research should focus on larger sample sizes and the optimal integration of VR with traditional teaching methods to fully explore its potential in medical education.

VR in Design and Architecture

Wang and Wang, (2024) explore the application of Virtual Reality (VR) and Computer-Aided Design (CAD) technologies in the design and development of cultural creativity products, addressing the challenge of inefficiencies in traditional design methods. Using an innovative 3D modelling approach that integrates VR and CAD, the study analyses optimization processes, structural refinement, and hierarchical simplification algorithms, demonstrating high accuracy (above 90%) and improved efficiency in product design. Despite its

advantages, challenges such as cost and resource limitations remain, with future research suggested to focus on integrating artificial intelligence and emerging digital technologies to further enhance design processes and accessibility.

Zhang, (2024) explores the application of virtual reality (VR) technology in architectural landscape design, highlighting its advantages in enhancing visualization, interactivity, and user experience but noting its limited adoption in real projects. The study employs VRML-based 3D modeling and scene optimization using LOD hierarchical detail algorithms, complemented by user experience evaluations through surveys, indicating that VR significantly improves design efficiency and user satisfaction. However, the research identifies a gap in widespread implementation and proposes further development of VR integration in landscape design, aiming to enhance digital transformation and accessibility in the industry.

Conclusion

In conclusion, Augmented Reality (AR) and Virtual Reality (VR) have undergone remarkable development, with state-of-the-art innovations improving their capabilities and broadening their use in multiple industries. These technologies are transforming user experiences and creating new opportunities in fields such as healthcare, education, and entertainment, among others. As advancements in hardware, software, and user engagement progress, the future of AR and VR holds the potential for even more groundbreaking innovations, further embedding these immersive technologies into daily life and facilitating significant transformations across various sectors.

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